Gonzalo Sebastián Bueno Bustíos

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Education

London School of Economics and Political Science — London, UK

MSc Economics (Distinction, 11/150 top 7%) — *2025–2026*

- Selected Coursework: Microeconomics (5/166 top 3%), Macroeconomics (9/169 top 5%), Econometrics (24/168 top 14%), International Economics (4/14 top 30%).
- Essay: "Is Peru's FX Intervention Optimal? A Welfare Analysis of the 2021 Depreciation" (Grade: Distinction 75/100. Supervisor: Dmitry Mukhin).
- Scholarship awarded by the Banco Central de Reserva del Perú.

Banco Central de Reserva del Perú — Lima, Perú

LXVIII Advanced Economics Extension Course — *2021*

- Postgraduate-level course: Monetary Policy, New Keynesian Models, International Macroeconomics, Numerical Methods for Economics, Economic Forecasting Methods, Financial Programming.
- Fifth place among thirty participants.

Universidad del Pacífico (UP) — Lima, Perú

B.S. in Economics, School of Economics and Finance — *2015–2020*

- Top quarter of cohort (17th of 68). Top fifth across all students (2017, 2018); Top tenth (2019, 2020).
- Letter of Recognition for Academic Merit awarded by the University President.
- Selected Coursework: Calculus, Linear Algebra (Eigenvalues, Eigenvectors), Differential Equations, Topology and Optimization Methods, Statistics I–II, Microeconomics I–II, Macroeconomics I–III, Econometrics I–II.

Research Experience

Banco Central de Reserva del Perú (BCRP) — Lima, Perú

Macroeconomic Modeling Specialist — *Jul 2024 – Present*

- Produce macroeconomic projections and risk scenarios for the monetary policy decision using the BCRP's Quarterly Projection Model (QPM). Built the first object-oriented Python prototype of the forecasting pipeline (forecast automation, impulse-response analysis, scenario management around a DSGE core with unanticipated shocks), since taken forward as the basis for the ongoing migration from MATLAB.
- Prepared briefing notes on international shocks and the external environment presented by the BCRP's Chief Economist at BIS meetings (Consultative Council for the Americas; CGMP), and a note for the BCRP's Governor on the natural rate of interest for commodity-exporting economies.
- Delivered applied projects for the quarterly *Reporte de Inflación*, published as boxes (*recuadros*): inflation expectations feedback dynamics, non-linear terms-of-trade effects on GDP, and potential output measurement.
- Monthly GDP nowcasting using mixed-frequency methods for a real-time signal in the forecasting round.
- *Study leave Aug 2025 – Jul 2026.*

Price Statistics Specialist — *May 2021 – Jun 2024*

- Designed and built the BCRP's first daily item-level CPI (∼500,000 price observations): web-scraping pipelines and ML-based text classification into COICOP categories across 500+ products. Adopted for real-time inflation monitoring and policy analysis.
- Analyzed supermarket price microdata on price-setting behavior and its gaps with official inflation statistics; reported findings directly to the BCRP Board. Presented at the 12th Biennial IFC–BIS Conference (Basel, 2024) and published in the *IFC Bulletin*.
- Found that through the 2020–2024 inflationary episode firms adjusted prices mainly on the frequency margin while average change size stayed flat, evidence of state-dependent rather than time-dependent pricing. Developed three working papers based on this research.
- Served on the BCRP Data Analytics Committee, working with the General Manager to promote data science tools across departments.

Red de Estudios para el Desarrollo (REDES) Lima, Perú

Research Intern *Mar 2020 – Dec 2020*

- Literature reviews and empirical analysis on education, health, and macroeconomic policy. Drafted policy briefs and background notes; co-authored a published report on the effects of COVID-19 on education.
- Co-authored newspaper articles on economic policy with Oswaldo Molina for *El Comercio*.
- Supported Asbanc initiatives on the Reactiva Perú liquidity program; prepared presentations for external institutions.

Publications and Research

Published

- Bueno, G. (2026). "Inflation and Price Dynamics Analysis using High-Frequency Data from Supermarkets." *IFC Bulletin*: "Statistics and beyond: new data for decision making in central banks," Bank for International Settlements.
- Bueno, G. and Olivera, L. (2023). "Patrones de precios de alimentos procesados." *Revista Moneda*, Banco Central de Reserva del Perú, 196, pp. 45–55.
- Bueno, G. and Hernández, G. (2022). "El efecto de los estereotipos de género y la maternidad en la brecha de género en puestos de liderazgo formales en el sector privado: una mirada experimental." In *Economía aplicada: ensayos de investigación económica 2020*, Universidad del Pacífico.

Work in Progress

Working papers

- "What Does FX Intervention Buy? A Quantitative Evaluation of Peru's Managed Float." Based on LSE MSc essay. Draft available.
- "Traspaso de los términos de intercambio hacia el tipo de cambio y su evolución en Perú." [Terms-of-trade pass-through to the exchange rate and its evolution over time in Peru]. With Flavio Abanto, Guillermo Ferreyros and Youel Rojas (BCRP). In preparation.
- "Analyzing Peruvian Inflation Through the Lens of Menu Cost Model." With Luis-Gonzalo Llosa (BCRP, UP). In preparation.
- "Do Firms Adjust Prices through Frequency or Magnitude? Granular High-Frequency Evidence from Peru's Inflationary Episode (2020–2024)." Draft available.
- "From Daily Prices to Inflation Trends: Constructing a Daily Consumer Price Index for Peru." Draft available.

Policy articles

- "Discusión de teoría de política de intervención cambiaria y hechos estilizados para Perú." [Discussion of the theory for foreign exchange intervention and stylized facts for Peru]. *Revista Moneda*. Draft submitted.
- "Crecimiento económico y tasa natural de interés" [Economic growth and the natural interest rate]. *Revista Moneda*. With Luis Zapata and Youel Rojas (BCRP). Draft available.

Conferences and Seminars

- "Is Peru's FX Intervention Optimal? A Welfare Analysis of the 2021 Depreciation." BCRP Internal Research Seminar (virtual), Lima, 2026.
- "From Daily Prices to Inflation Trends: Constructing a Daily Consumer Price Index for Peru." BCRP Internal Research Seminar (virtual), Lima, 2025.
- "Análisis de la inflación y la dinámica de precios utilizando datos de alta frecuencia de supermercados: evidencia para el caso de Perú" [Inflation and price dynamics analysis using high-frequency supermarket data: evidence from Peru]. Seminario de Economía, Banco Central de la República Dominicana (virtual), 2025.
- "From Daily Prices to Inflation Trends: Comparing Stylized Facts in Supermarket Data with Official Inflation Statistics." XLII Encuentro de Economistas del BCRP, Lima, 2024.
- "Inflation and Price Dynamics Analysis using High-Frequency Data from Supermarkets." 12th Biennial IFC Conference, Basel, Switzerland, 2024.
- "Extracción, clasificación y procesamiento de datos de alta frecuencia de supermercados" [Extraction, classification, and processing of high-frequency supermarket data]. XLI Encuentro de Economistas del BCRP, Lima, 2023.

Teaching Experience

Banco Central de Reserva del Perú Lima, Perú
Teaching Assistant, LXXII Advanced Economics Extension Course (Monetary Policy) *Mar 2025*

Universidad del Pacífico Lima, Perú
Teaching Assistant *May 2018 – Jun 2023*

- Five years across six courses: International Macroeconomics (created materials for practice sessions), Econometrics I, Mathematics III (Linear Algebra, Differential Equations), Mathematics IV (Topology, Optimization), Macroeconomics II (Monetary), Introduction to Microeconomics.

Awards & Honors

BCRP Study Leave Scholarship (MSc Economics, LSE)	2025
Employee of the Year, Information and Economic Analysis Division, BCRP	2022
Second Place, BCRP Innovation Award	2022
Letter of Recognition for Merit, President, Universidad del Pacífico	2017

Research Software and Tools

EconJarvis | *Research infrastructure for AI-assisted economics* 2026 – Present

- Local-first research knowledge system combining reusable academic document corpora, keyword and semantic retrieval, and page-level citations, exposed to AI assistants through a Model Context Protocol (MCP) interface. Original documents and revision histories are preserved; separate project workspaces hold evidence, hypotheses, and methodological decisions. Working prototype under active development.

BCRP Corpus | *AI-assisted research and information retrieval for central banking* 2026

- Built a page-grounded retrieval-augmented generation (RAG) system over BCRP publications: 474 working papers, 788 *Revista Moneda* articles and 89 inflation reports (1,441 indexed records, >30,000 pages, ∼97,000 searchable passages), with an automated pipeline for acquisition, metadata extraction, OCR and incremental updates.
- Implemented multilingual hybrid retrieval (BM25, semantic embeddings, reciprocal rank fusion, cross-encoder reranking), exposed to AI assistants through a Model Context Protocol (MCP) server for literature discovery and evidence-based synthesis.
- Designed a provenance framework linking every retrieved passage to its original PDF page, publication date and extraction warnings, preserving the historical information set available at each date.
- Evaluated lexical, semantic and hybrid retrieval and agentic evidence-discovery workflows on citation accuracy, coverage and computational cost. Accompanying paper on the potential and limits of AI-assisted research in central banking: in preparation.

Technical Skills & Languages

Programming: Python (NumPy, Pandas, Scikit-learn, Scrapy/BeautifulSoup), R, Stata, Dynare, MATLAB, EViews, Git, LaTeX
AI-assisted research tooling: LLM agent workflows for design, implementation and review (Claude Code, Codex); RAG and hybrid retrieval (BM25, embeddings, reranking); Model Context Protocol
Methods: DSGE estimation, time series econometrics, ML text classification, web scraping, high-frequency data analysis, nowcasting.
Languages: Spanish (native), English (fluent), French (basic, A2)